

2025 AMC 10A Problem 23 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 23. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 23

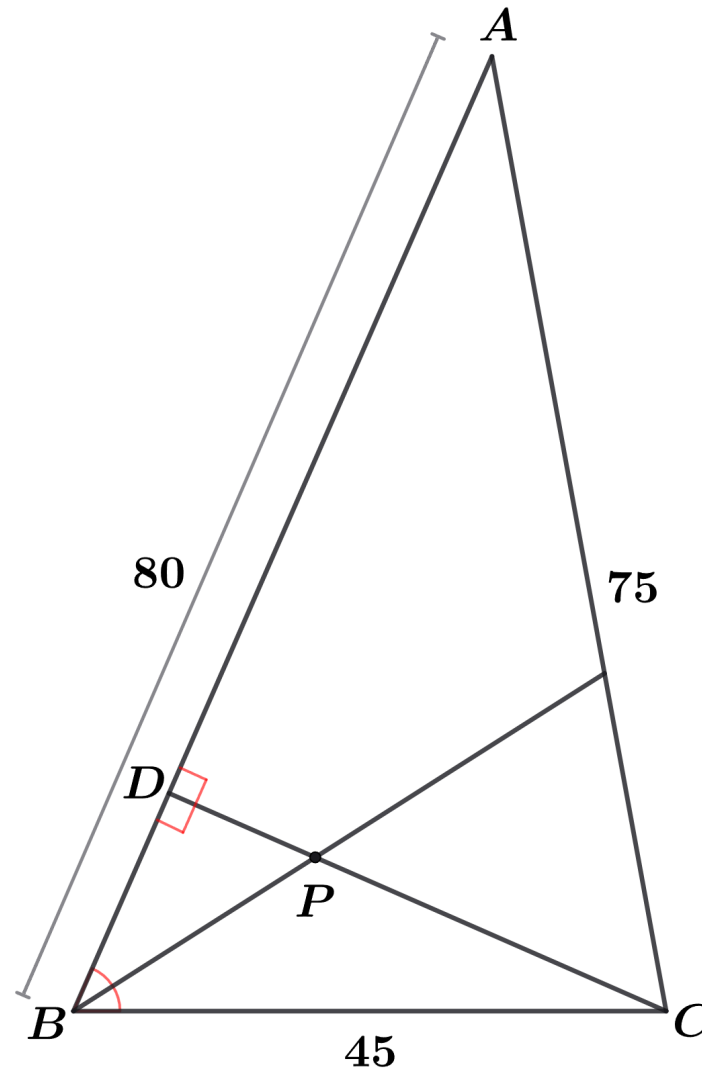
**Problem:**

Triangle $\triangle ABC$ has side lengths $AB = 80$, $BC = 45$, and $AC = 75$. The bisector of $\angle B$ and the altitude to side $\overline{AB}$ intersect at point P . What is BP ?

Answer Choices:

- A. 18
- B. 19
- C. 20
- D. 21
- E. 22

Solution:



Let D be the foot of the altitude from C to $\overline{AB}$, and let P be the intersection of BD (the bisector of $\angle B$) with $\overline{CD}$. We find AD and BD using the Pythagorean theorem.

From right triangles ADC and BDC :

$$AC^2 = AD^2 + CD^2,$$

$$BC^2 = BD^2 + CD^2.$$

From above, we get:

$$AC^2 - BC^2 = AD^2 - BD^2 = (AD - BD)(AD + BD).$$

$$\begin{aligned} AD + BD = AB = 80, AC = 75, BC = 45, &\implies (75^2 - 45^2) = 80(AD - BD) \\ &\implies (75 - 45)(75 + 45) = 80(AD - BD) \\ &\implies 30 \cdot 120 = 80(AD - BD) \\ &\implies AD - BD = 45. \end{aligned}$$

Solving

$$\begin{cases} AD + BD = 80, \\ AD - BD = 45, \end{cases}$$

we get

$$AD = \frac{125}{2}, BD = \frac{35}{2}.$$

We find CD by applying the Pythagorean theorem in $\triangle BDC$.

$$CD = \sqrt{BC^2 - BD^2} = \sqrt{45^2 - \left(\frac{35}{2}\right)^2} = \sqrt{\frac{6875}{4}} = \frac{25}{2}\sqrt{11}.$$

We use the Angle Bisector Theorem in $\triangle BCD$ to find DP . Since BP bisects $\angle B$,

$$\frac{DP}{PC} = \frac{BD}{BC} = \frac{\frac{35}{2}}{45} = \frac{7}{18} \Rightarrow DP = \frac{7}{25} CD.$$

Hence,

$$DP = \frac{7}{25} \cdot \frac{25}{2} \sqrt{11} = \frac{7\sqrt{11}}{2}.$$

Now, we can again use the Pythagorean Theorem in $\triangle BDP$ to find BP .

$$BP^2 = BD^2 + DP^2 = \left(\frac{35}{2}\right)^2 + \left(\frac{7\sqrt{11}}{2}\right)^2 = \frac{1225}{4} + \frac{539}{4} = \frac{1764}{4} = 441.$$

so $BP = \boxed{\text{(D) } 21}$.

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